Clustering Mythic Themes: A Simple Machine Learning Experiment on Greek and Roman Myths

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1 Introduction

For this project, I wanted to see what would happen if I treated Greek and Roman myths as data. Instead of writing a traditional essay, I built a small machine learning pipeline—a computer method that looks for patterns in data—to explore how a model would group mythic scenes from Homer's *Odyssey* and Ovid's *Metamorphoses* (Hom. Od.; Ov. Met. 6). My main question was whether the clusters found by the model would match the themes we talk about in class—especially xenia, hubris, fate, divine punishment, loyalty, and gender roles—or whether it would group stories in ways that surprised me.

2 Data and Annotation

I started by building a small corpus of passages rather than trying to analyze whole books. I chose fifteen short scenes: thirteen from the *Odyssey* in the Butler translation and two from *Metamorphoses* Book 6 in the Brookes More translation (*Arachne* and *Niobe*). The *Odyssey* passages include the Cyclops episodes, Circe, Helios' cattle, the Sirens, Scylla and Charybdis, Telemachus confronting the suitors, the bow contest, the slaughter of the suitors, Nausicaa, Calypso, Eumaeus, and Argos (Hom. Od. 1.368–420; 5.118–152; 6.149–185; 9.272–307, 9.500–542; 10.229–243; 12.37–54, 12.73–110, 12.339–365; 14.50–90; 17.290–327; 21.118–376; 22.8–41; Ov. Met. 6.1–53; 6.267–303). For each passage, I wrote a one-sentence "lesson" in my own words and tagged it with themes like xenia, justice, hubris, fate, divine_punishment, loyalty, and gender_roles. These tags became my human baseline for what I thought each scene was "about."

3 Methods

On the technical side, I used TF-IDF, a basic text technique that turns each passage into numbers by giving higher scores to words that are frequent in that passage but not common everywhere. In practice, this meant each passage became a vector, a long list of numbers, and all the vectors together formed a matrix, a big table, where rows were passages and columns were words. Once I had this matrix, I used k-means clustering, a simple algorithm that groups similar rows together, to sort the passages into a chosen number of groups. I wrote helper functions in Python (`build_tfidf` and `run_kmeans`) so I could change the number of clusters k, the number of groups I ask the algorithm to make, easily, and re-run everything. I started with a tiny dataset and a small k, then expanded the corpus up to fifteen passages and experimented with k = 3, 4, and 5.

4 Results

My main model choice was k = 4, because it gave clusters that were specific enough to interpret without collapsing into single scenes. Table 1 summarizes all fifteen passages, their works, my theme tags, and

their assigned cluster in the k = 4 model. Figure 1 shows a two-dimensional projection of the TF-IDF space, where each passage is represented as a point and colored by its k = 4 cluster. Table 2 compares the cluster assignments for each passage when I set k = 3, 4, and 5, so it is easier to see how changing k reshapes the groupings.

One cluster grouped the sea-trials—Helios' cattle, the Sirens, and Scylla and Charybdis—as scenes where leaders face impossible choices and partial losses under divine pressure (Hom. Od. 12.37–54, 12.73–110, 12.339–365). Another cluster tied together Circe's pigs, Niobe, Eumaeus, and Argos, mixing everyday loyalty and hospitality with temptation and downfall (Hom. Od. 10.229–243; 14.50–90; 17.290–327; Ov. Met. 6.267–303). A third cluster focused on household justice and hubris, bringing together Telemachus confronting the suitors, Odysseus' boast to the Cyclops, the bow contest, the slaughter of the suitors, and Arachne's challenge to Minerva as a parallel case of pride and punishment (Hom. Od. 1.368–420; 9.500–542; 21.118–376; 22.8–41; Ov. Met. 6.1–53). The final cluster links Polyphemus refusing proper hospitality, Nausicaa's cautious but generous welcome, and Calypso criticizing the other gods, which together raise questions about who sets the rules for xenia and how power and gender shape those rules for both mortals and immortals (Hom. Od. 5.118–152; 6.149–185; 9.272–307). In most cases, the clusters lined up well with my tags, but the model also forced me to think harder about why Arachne felt at home with household justice scenes or why Niobe sometimes stood out as an outlier when I changed k (Ov. Met. 6.1–53, 6.267–303).

Because the PCA plot compresses many TF-IDF dimensions down to two, some clusters appear closer together in Figure 1 than they are in the full high-dimensional space, but the color-coding still reflects the underlying k-means assignments.

Table 1. Passages, their works, my theme tags, and their cluster assignments in the k = 4 model.

	id	title	work	cluster_k4	tags
5	helios_cattle	The Slaughter of Helios' Cattle	Odyssey	0	disobedience, fate, leadership_failure
14	scylla_charybdis	Scylla and Charybdis Described	Odyssey	0	fate, leadership_failure, suffering
13	sirens_song	Circe Warns Odysseus About the Sirens	Odyssey	0	temptation, cunning, fate
9	niobe_hubris	Niobe's Fall and Transformation	Metamorphoses	1	hubris, grief, divine_punishment
11	argos_loyal_dog	Argos Recognizes Odysseus	Odyssey	1	loyalty, homecoming, neglect
4	circe_pigs	Circe Turns the Men into Swine	Odyssey	1	temptation, enchantment, cunning_vs_naivety
10	eumaeus_hospitality	Eumaeus Welcomes the Stranger	Odyssey	1	xenia, loyalty, class
8	arachne_hubris	Arachne Challenges Minerva	Metamorphoses	2	hubris, craft, divine_punishment
2	cyclops_boast_curse	Odysseus' Boast and the Cyclops' Curse	Odyssey	2	hubris, fate, consequences
12	penelope_bow_contest	The Contest of the Bow	Odyssey	2	justice, xenia, hubris, fate
7	suitors_slaughter	Odysseus Kills Antinous and Condemns the Suitors	Odyssey	2	justice, xenia, retribution
0	telemachus_suitors	Telemachus Confronts the Suitors	Odyssey	2	xenia, justice, coming_of_age
6	calypso_double_standard	Calypso Complains About the Gods	Odyssey	3	divine_power, fate, gender_double_standard
1	cyclops_violated_xenia	Polyphemus Rejects Hospitality	Odyssey	3	xenia, brutality, divine_law
3	odysseus_nausicaa	Odysseus Speaks to Nausicaa	Odyssey	3	xenia, cunning, gender_roles

Figure 1. Two-dimensional PCA projection of the TF-IDF vectors, with each passage colored by its k = 4 cluster.

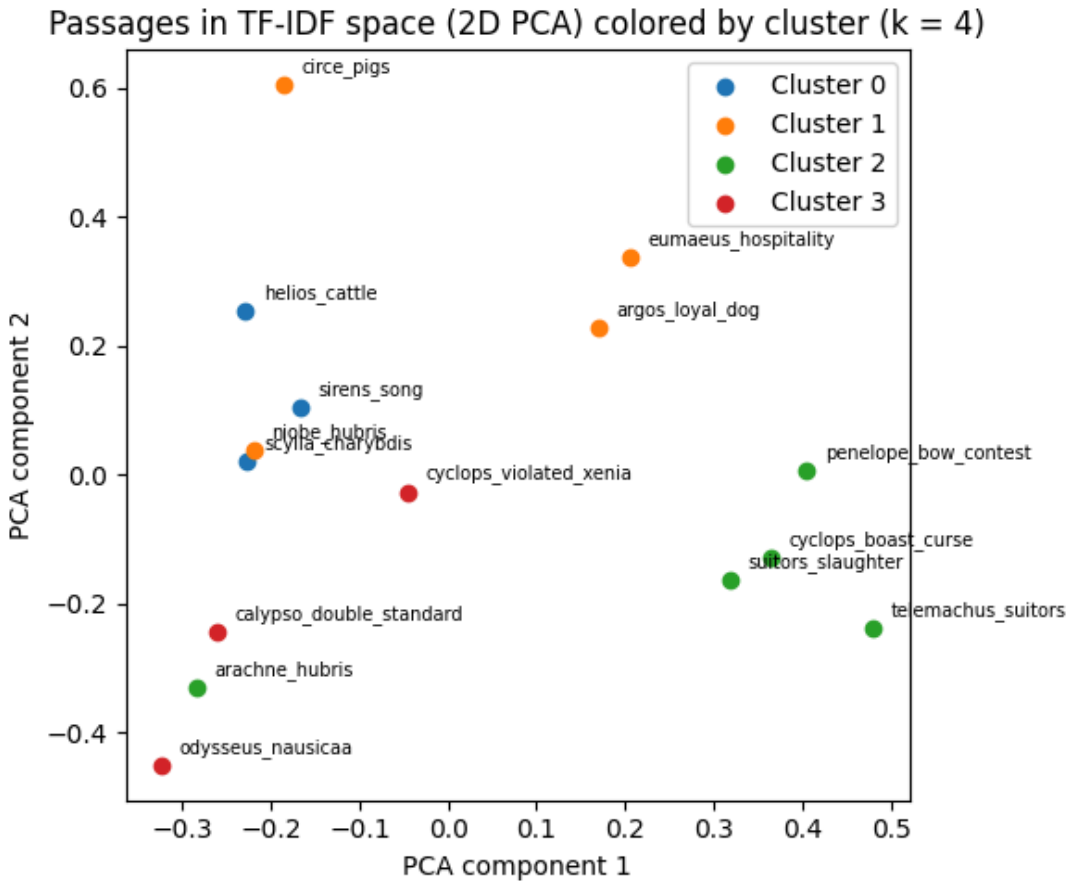

Table 2. Cluster assignments for each passage when k = 3, 4, and 5, showing how changing k reshapes the groupings.

	id	title	cluster_k3	cluster_k4	cluster_k5
8	arachne_hubris	Arachne Challenges Minerva	2	2	3
11	argos_loyal_dog	Argos Recognizes Odysseus	0	1	0
6	calypso_double_standard	Calypso Complains About the Gods	2	3	1
4	circe_pigs	Circe Turns the Men into Swine	0	1	0
2	cyclops_boast_curse	Odysseus' Boast and the Cyclops' Curse	1	2	2
1	cyclops_violated_xenia	Polyphemus Rejects Hospitality	0	3	1
10	eumaeus_hospitality	Eumaeus Welcomes the Stranger	0	1	0
5	helios_cattle	The Slaughter of Helios' Cattle	0	0	1
9	niobe_hubris	Niobe's Fall and Transformation	0	1	4
3	odysseus_nausicaa	Odysseus Speaks to Nausicaa	2	3	3
12	penelope_bow_contest	The Contest of the Bow	1	2	2
14	scylla_charybdis	Scylla and Charybdis Described	0	0	1
13	sirens_song	Circe Warns Odysseus About the Sirens	0	0	0
7	suitors_slaughter	Odysseus Kills Antinous and Condemns the Suitors	1	2	2
0	telemachus_suitors	Telemachus Confronts the Suitors	1	2	2

5 Discussion / Project Commentary

The hardest part of the project was the process, not the math. I had to figure out how to copy reliable passages from different translations, decide where scenes should start and end, and keep my data consistent. Since I am still relatively new to Python and to machine learning ideas like TF-IDF and k-means, I had to build the Google Colab notebook slowly: first just printing texts, then building the TF-IDF matrix, then running k-means, and finally wrapping it all in one analysis cell that printed clusters, top words, and tag summaries. I also had to accept that there is no single "correct" k. Instead, changing k changes the story the model tells, and part of my job was to explain why I found k = 4 the most useful for this particular set of passages.

6 Conclusion and Future Work

This project taught me that machine learning does not replace close reading; it gives me another way to interrogate my own assumptions. The model only sees word patterns in numerical form, but those patterns push me to justify why I think certain myths belong together under themes like xenia or hubris, and to notice when the computer sees a connection I might have missed. With more time, I would expand the corpus to include tragedy and more Ovid, and I would experiment with word embeddings (methods that represent words based on the company they keep, not just their counts) or softer clustering methods that allow passages to belong partly to more than one cluster. Even in this small form, though, the project helped me connect course content, my own thematic tags, and a simple ML model into one coherent experiment.

Works Cited

Homer. *The Odyssey*. Translated by Samuel Butler. Theoi Greek Mythology, https://www.theoi.com/Text/HomerOdyssey1.html.

Ovid. *Metamorphoses*. Translated by Brookes More. Theoi Greek Mythology, https://www.theoi.com/Text/OvidMetamorphoses6.html.